

Sustaining Juneau: A Policy-Driven Approach to Tourism Resilience

Summary

As a key tourist destination in Alaska, Juneau faces the challenge of balancing economic growth from tourism with environmental sustainability, leaving it with an urgent mission to optimize tourism policies by integrating economic, ecological, and social dimensions.

In this context, we propose a **Sustainable Tourism Optimization and Evaluation Model** for Juneau, with the aim of guiding policy decisions. Sustainability is categorized into three main dimensions: economic, ecological, and social. Due to the growing concern over the implicit costs of tourism, we also introduce an infrastructure health indicator. For each dimension, representative sub-indicators are selected, including tourism surplus, carbon dioxide emissions, glacier melt, resident satisfaction, and infrastructure health.

These indicators are interdependent and linked through various variables. To quantify the health of each dimension, we normalize the scores on a 0-1 scale, where higher scores represent better performance. **The Analytic Hierarchy Process (AHP)** is applied to assign weights to the four major categories, which results in an overall sustainability score.

In order to support policy development, we establish function relationships between controllable variables and sub-indicators using historical data. This is achieved through **nonlinear regression** and **logistic regression**. The function relationships are informed by key economic and sociological theories, including price elasticity of demand, diminishing marginal returns, and congestion effects.

After establishing these relationships, we design multiple policy scenarios, each with three adjustment dimensions for each controllable variable. These scenarios decompose the government's long-term policy goals into an annual incremental progression. The future 20-year sustainability scores are predicted by solving these progressions. A comparative analysis of 81 potential policy combinations is conducted to derive corresponding policy recommendations.

Based on the **policy scores** and **sensitivity analysis**, our recommendations for Juneau are as follows: implement tourist diversion measures, maintain stable tourist numbers, increase environmental investment, gradually raise tax rates to increase government revenue, and reduce excessive infrastructure development.

Finally, we adapt the model for application to tourism cities in developing countries (e.g., Angkor Wat) and cities with emerging tourism industries (e.g., Khorog). The model demonstrates strong explanatory power across different types of cities, providing valuable, adaptable policy recommendations based on a combination of sensitivity analysis and sustainability scores.

Keywords: Sustainable Tourism Optimization and Evaluation Model; Analytic Hierarchy Process (AHP); Nonlinear Regression Model; Logistic Regression Model

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1 Introduction

1.1 Problem Background

Juneau, Alaska, with a population of approximately 30,000, has become a popular hub for tourists, particularly cruise passengers. In 2023, the city welcomed 1.6 million visitors, generating over \$375 million in revenue. However, the influx of tourists is placing immense pressure on both the local infrastructure and its natural environment. The Mendenhall Glacier, once a pristine natural attraction, is now retreating at an alarming rate due to climate change, compounded by tourism.

A recent study, published in *Nature Communications*, reveals that the Juneau Icefield, home to the Mendenhall Glacier, has been losing volume at an accelerating pace since 2005. Between 2015 and 2019, the glacier's retreat was five times faster than between 1979 and 1990. This rapid melt is primarily driven by rising equilibrium line altitudes and feedback mechanisms that accelerate the melting process, pushing the glacier towards an irreversible tipping point. If current trends continue, the glacier could be unrecognizable by 2050.

As the glacier recedes, the city faces a critical question: How can Juneau sustain its thriving tourism industry, heavily reliant on this natural wonder, while also ensuring its preservation for future generations? This challenge highlights the need for a sustainable tourism model—one that balances economic growth, environmental stewardship, and social well-being.

1.2 Restatement of Problem

We are required to develop effective policies and management approaches for tourist destinations to adapt to long-term changes. Our goal is to strike a balance between regulating visitor flow, safeguarding the environment, promoting sustainable infrastructure, and addressing the concerns of local communities alongside the impact of tourism. Therefore, we believe that the whole work can be divided into 4 problems:

- ⊙ **Problem 1:** Build a model for sustainable tourism in Juneau which integrates factors like visitor numbers, total revenue, infrastructure investments, and environmental impacts, optimizing these elements while adhering to constraints like infrastructure capacity and environmental sustainability.
- ⊙ **Problem 2:** Apply the model to other destinations affected by over-tourism, analyzing regional characteristics and adjusting the model to meet the specific needs of each area.
- ⊙ **Problem 3:** Generalize our model to promote lesser-visited attractions while ensuring a balanced tourism.
- ⊙ **Problem 4:** Provide a concise memo to the Juneau Tourist Council summarizing the model's predictions, the impact of various measures, and recommendations for optimizing tourism management.

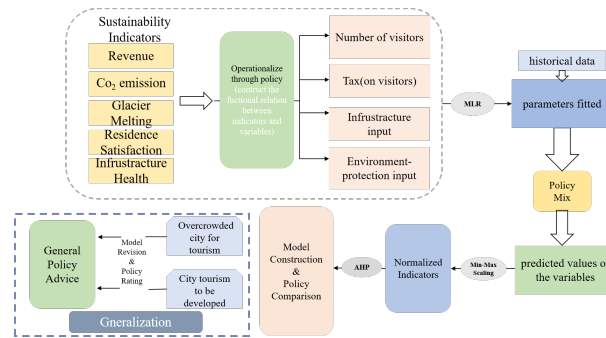


Figure 1: Flow Chart

2 Assumptions

- ◇ **Assumption 1:** The government can control four variables: the annual tourism tax rate, tourist number cap, infrastructure investment, and ecological protection investment.
 - ↪ **Justification:** These variables are directly controlled by government policies, such as fiscal policies for tax rates, visa and ticketing policies for tourist caps, and budget allocations for infrastructure and environmental investments. For simplicity, only these four variables are included, as they cover the key economic, ecological, and social dimensions.
- ◇ **Assumption 2:** Once the annual tourist number cap is set, the actual tourist number is assumed to be equal to the cap.
 - ↪ **Justification:** In popular tourist destinations, demand often exceeds capacity, so the actual number of tourists will approach the cap once it is established.
- ◇ **Assumption 3:** Changes in the tourism tax rate, tourist number, infrastructure investment, and ecological protection investment are independent of each other.
 - ↪ **Justification:** These variables are set independently by government policies—fiscal policies for tax rates, regulations for tourist caps, and budget decisions for infrastructure and ecological investments. While interactions may exist in the long term, they are assumed to be independent in the short term for model simplification.
- ◇ **Assumption 4:** These variables are set independently by government policies—fiscal policies for tax rates, regulations for tourist caps, and budget decisions for infrastructure and ecological investments. While interactions may exist in the long term, they are assumed to be independent in the short term for model simplification.
 - ↪ **Justification:** Tourism revenue depends on tourist numbers and tax rates, while CO2 emissions and glacier melt are influenced by tourist activities, mitigated by ecological investments. Resident satisfaction is affected by congestion and infrastructure capacity, and infrastructure health is determined by investment levels and tourist numbers. Unforeseen events are not considered for simplification.

3 Notations

Table 1: Important Notations.

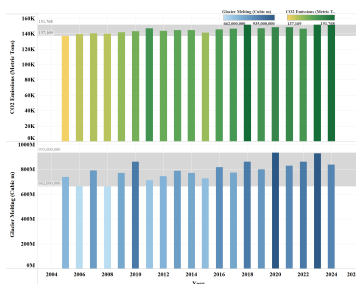
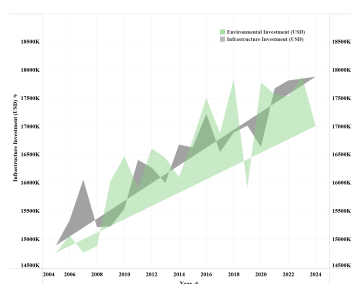
Symbol	Description	Unit
N_v	number of tourists	Individual
T_v	tourism tax rate	%
C_{env}	environmental investment	USD
C_{infra}	infrastructure investment	USD
R	tourism surplus	USD
CO_2	carbon dioxide emissions	10,000 tons
Glacier Melting	glacier melting volume	cubic km
Resident Satisfaction	resident satisfaction	%
IHI	infrastructure health index	USD/person
Wastewater Production	wastewater production volume	cubic meters
P	promotional investment	USD
Grass Degradation	grass degradation level	square meters

4 Data Description and Visualization

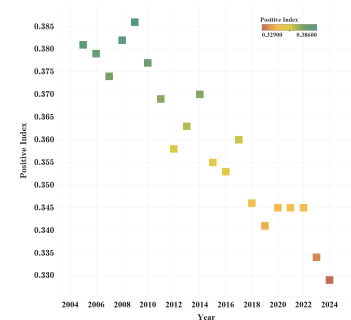
We primarily collected data on annual tourism revenue, Alaska's tourism-related CO2 emissions, glacier degradation volume, government taxes and reinvestments (such as environmental protection investments and infrastructure development), and resident satisfaction in Juneau City for parameter fitting of the model. The data sources are shown in Table 2.

Table 2: Data Sources for Juneau Tourism Analysis

Data Type	Data Source
Juneau Municipal Financial Statement	https://juneau.org/
Residence Satisfaction Rate	Juneau Tourism Survey
Juneau's Annual Carbon Footprint	https://www.eia.gov/
Mendenhall Glacier Melting Volume	https://static-content.springer.com

(a) CO₂ Emission and Glacier Melting Volume

(b) Environment vs. Infrastructure Investment



(c) Residence Satisfaction

By visualizing and analyzing historical data, we observe that the carbon emissions in Juneau have remained relatively stable with a slight upward trend, potentially due to the intervention of existing environmental policies. In contrast, the increase in glacier melt is more pronounced. Environmental investment exhibits greater fluctuations compared to infrastructure spending, reflecting the distinct characteristics of these two types of policies. Additionally, resident satisfaction with the local tourism industry shows a declining trend.

5 Sustainable Tourism Optimization and Evaluation Model

Our primary task involves developing a model to assess the sustainable development level of tourism in the Juneau region on an annual basis, and to project changes in the future sustainability of tourism under different policy scenarios, ultimately providing reasonable policy recommendations.

5.1 Evaluation Model

Sustainable development is typically assessed through three dimensions: social, economic, and ecological. For the city of Juneau, we selected five representative indicators for evaluation: annual tourism surplus, annual CO2 emissions, annual glacier melt volume, resident satisfaction, and infrastructure health index. In addition, considering that the model is intended to support government policy regulation, we included four adjustable policy-related independent variables: annual tourism tax rate, visitor number cap, infrastructure investment, and ecological environmental protection funding. Based on the assumption that these four variables are independent and can uniquely determine the five evaluation indicators, we treat the five annual indicators as functions of these four independent policy variables.

5.1.1 Economic Dimension Evaluation Indicators

We define R as the government's tourism surplus, which serves as the economic dimension evaluation indicator for Juneau's tourism economic development.

Firstly, tourism surplus is the difference between revenue and costs ($P = I - C$), where revenue is determined by the total spending of tourists, and the maintenance costs of the tourism industry are composed of investments in ecological environmental protection and infrastructure development. Given that an increase in tax rates reduces individual tourist spending, we refer to the Price Elasticity of Demand (PED) model, which discusses the sensitivity of tourist numbers to price changes and directly impacts tourism revenue.

When T_v changes, its effect on tourist numbers should be considered. Typically, it is assumed that the tax rate has a negative impact on tourist demand, meaning that higher taxes lead to fewer tourists. The demand elasticity model can be expressed as follows:

$$R = b \cdot (1 + T_v) \cdot N_v \cdot e^{-a \cdot T_v^2} - C_{\text{infra}} - C_{\text{env}} \quad (1)$$

If the tax rate increases, the overall revenue will also increase, and this direct effect is reflected in the term $1 + T_{v1} + T_v$. The factor $e^{-a \cdot T_v^2}$ is included to simulate the suppressive effect of the tax rate on tourist numbers and spending. While an increase in the tax rate may lead to higher revenue (as expressed by the term $b(1 + T_v)$), high taxes will also reduce tourist numbers or decrease tourists' willingness to spend. This suppressive effect is described using an exponential function, where N_0 represents the baseline number of tourists, and a is the demand

elasticity coefficient, indicating the sensitivity of tourist numbers to changes in the tax rate. The exponential decay model effectively captures this nonlinear and marginally diminishing effect, which aligns with the economic concept of demand elasticity.

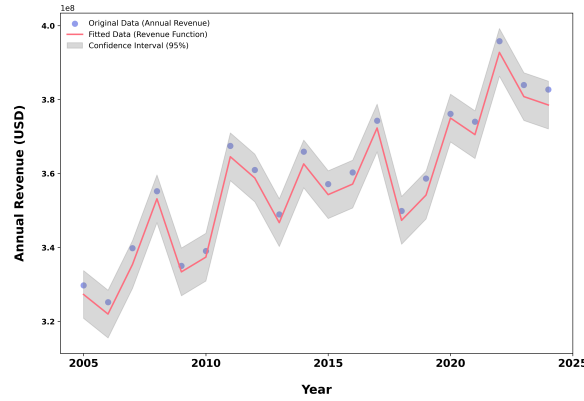


Figure 2: Simulation for R.

Thus, the preliminary functional relationship between net income and several basic variables can be expressed as:

$$R = 250 \cdot (1 + T_v) \cdot N_v \cdot e^{-0.01 \cdot T_v^2} - C_{\text{infra}} - C_{\text{env}} \quad (2)$$

5.1.2 Ecological Dimension Evaluation Indicators

The ecological dimension assesses the impact of tourism on the environment, particularly the extent of natural resource consumption, ecosystem degradation, and environmental pollution. The carbon footprint of tourism is a globally recognized environmental issue, as the tourism industry not only directly causes emissions during transportation (such as air and maritime travel) but also indirectly generates emissions through accommodation, dining, and other activities. These emissions represent the hidden costs of sustainable tourism development. In the context of Juneau, we propose that annual carbon dioxide (CO₂) emissions and annual glacier melting volume be selected as the primary ecological evaluation indicators.

(1) CO₂ emissions

Juneau is a typical tourist destination predominantly known for its natural landscapes, especially glaciers, and the melting of glaciers and carbon emissions are key factors affecting the ecological sustainability of the region. As tourism mainly relies on maritime cruises, which have high carbon emissions, assessing CO₂ emissions is particularly crucial. Simultaneously, glacier melting directly reflects the close link between tourism and climate change, especially considering that glaciers in Juneau serve as a major attraction.

This design aligns with the real-world situation where, with relatively low per capita environmental investment (such as the lack of widespread clean energy), the reduction effect is significant. However, when per capita environmental investment is higher, the marginal return on additional investment may face technological constraints, leading to diminishing effects on emission reductions.

There is a diminishing marginal benefit of environmental investment in reducing carbon

emissions. That is, increasing environmental investment significantly reduces emissions in the early stages, but after a certain threshold, the reduction effect weakens.

We introduce a nonlinear relationship $\left(\frac{C_{env}}{N_v}\right)^{\gamma_1}$ in the formula to reflect this diminishing effect. When $\gamma_1 < 0$, it indicates that the marginal reduction in emissions from environmental investment gradually weakens as investment increases. Basic human activities generate a fixed amount of CO₂ emissions, which we denote as B_1 , thus leading to the following model expression:

$$CO_2 = a_1 \cdot \left(\frac{C_{env}}{N_v}\right)^{\gamma_1} + B_1 \quad (3)$$

The parameter values were obtained by fitting historical carbon emission data. Specifically, $a_1 = 250,000$, $\gamma_1 = -\frac{1}{3}$, and $B_1 = 30,000$.

The value $a_1 = 250,000$ indicates that when per capita environmental investment is at a unit level (i.e., $\frac{C_{env}}{N_v} = 1$), the emission reduction effect of environmental investment can reduce CO₂ emissions by 250,000 tons. This value reflects the high efficiency of environmental funds in the tourism industry of Juneau.

The parameter $\gamma_1 = -\frac{1}{3}$ indicates that the emission reduction effect decreases as per capita environmental investment follows a cubic root decline.

The value $B_1 = 30,000$ tons represents the CO₂ emissions that will still result from tourism-related uncontrollable activities (such as basic tourist activities and transportation infrastructure needs), even with very high per capita environmental investment.

The final function relationship is expressed as:

$$CO_2 = 250000 \cdot \left(\frac{C_{env}}{N_v}\right)^{-1/3} + 30000 \quad (4)$$

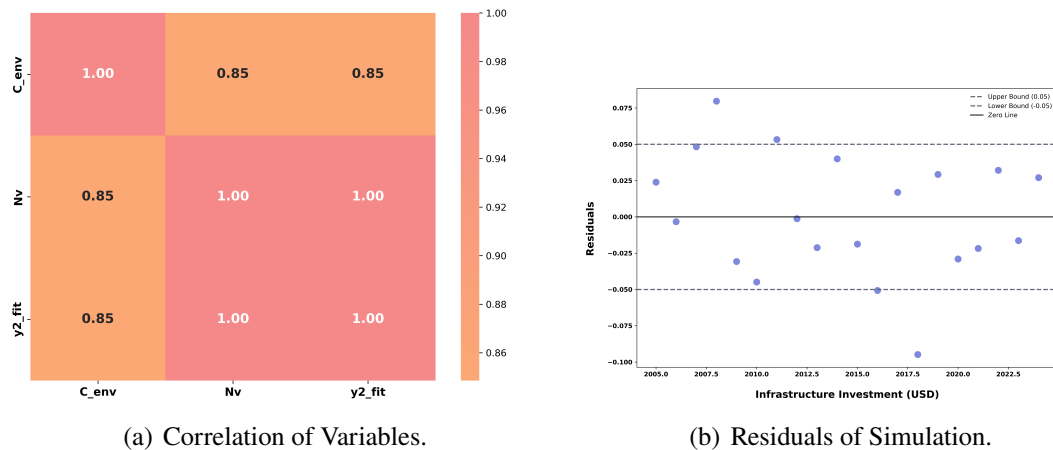


Figure 3: Fitting Accuracy Analysis(CO₂).

(2) Glacier Melting

The function for glacier melting is constructed similarly to the CO₂ emission model. Glacier melting is influenced by both environmental investment and the number of tourists. The higher

the per capita environmental investment, the better the suppression of human activities' impact on glacier melting, leading to a reduction in melting. However, considering the diminishing marginal benefits, γ_2 is set as a negative value. The fixed melting volume B_2 reflects the uncontrollable natural melting due to global warming.

$$\text{Glacier Melting} = a_2 \cdot \left(\frac{C_{\text{env}}}{N_v} \right)^{\gamma_2} + B_2 \quad (5)$$

The parameters are $a_2 = 1,000,000,000$, $\gamma_2 = -\frac{1}{4}$, and $B_2 = 200,000,000$.

We consider a_2 as a scaling factor used to quantify the total effect of per capita environmental investment $\frac{C_{\text{env}}}{N_v}$ on the mitigation of glacier melting. It amplifies the effect of unit per capita environmental investment to match the magnitude of glacier melting (such as in cubic meters or glacier volume units). Since glacier melting volumes are usually very large (for instance, the volume of ice melted annually can range from tens of millions to billions of cubic meters), a large coefficient a_2 is needed to match the actual scale of the data.

The parameter $\gamma_2 = -\frac{1}{4}$ indicates that the mitigation effect on glacier melting follows a quartic root relationship with per capita environmental investment.

The value $B_2 = 200,000,000$ represents the natural annual glacier melting volume of approximately 200 million cubic meters, even in the absence of tourist activity or environmental interventions.

The final function relationship is expressed as:

$$\text{Glacier Melting} = 1000000000 \cdot \left(\frac{C_{\text{env}}}{N_v} \right)^{\gamma_2} + 200000000 \quad (6)$$

5.1.3 Social Dimension Evaluation Indicators

The social dimension primarily focuses on the impact of tourism on the local residents' quality of life and social stability, measuring the overall perception of residents regarding the various effects brought by tourism, including economic benefits, environmental protection, and traffic pressure. In the case of Juneau, resident satisfaction is the most direct and representative social evaluation indicator. As the number of tourists increases and tourism develops, residents may experience issues such as overcrowding, resource scarcity, and higher taxes, which could lead to a decrease in support for tourism. Therefore, this indicator reflects the level of social acceptance of tourism and its potential for sustainable development.

Through a survey on tourism in Juneau, we obtained annual social satisfaction data. We believe that the resident satisfaction index is influenced by three factors: tax rate, tourist numbers, and infrastructure investment. The tax rate term ($-\gamma_4 T_v$) reflects the negative impact of taxes on resident satisfaction. A higher tax rate reduces the direct economic benefits that residents receive from tourism, leading to a decrease in satisfaction. An increase in tourist numbers leads to overcrowding and resource shortages, which also results in lower satisfaction. On the other hand, increased infrastructure investment improves the development of urban infrastructure, enhancing the convenience of residents' lives, which in turn raises satisfaction.

We examined the Crowding Effect Theory from sociology: when the demand for resource use exceeds the available supply capacity, a crowding effect occurs, leading to residents' dissatisfaction with resources and public services. This is reflected in $\frac{\gamma_3 N_v}{\text{Capacity}_{\text{social, base}} + \sigma_1 C_{\text{infra}}}$ in the

formula, where an increase in N_v (tourist numbers) intensifies resource competition, directly lowering resident satisfaction. Social carrying capacity and infrastructure investment, denoted as $\text{Capacity}_{\text{social, base}} + \sigma_1 C_{\text{infra}}$, which determines the supply capacity of resources. When N_v exceeds the available supply capacity, the fractional value in the formula rises, resulting in a decrease in the overall satisfaction index. Finally, we use the Sigmoid function to map the complex linear effects to the $[0, 1]$ range. The model can finally be expressed as follows:

$$\text{Residence Satisfactory} = \frac{1}{1 + e^{-\left(1 - \frac{\gamma_3 N_v}{\text{Capacity}_{\text{social, base}} + \sigma_1 C_{\text{infra}}} - \gamma_4 T_v\right)}} \quad (7)$$

Through nonlinear regression, we solved for the parameters: $\gamma_4 = 0.5$, capacity = 5.5, $\sigma = 1$, and $\gamma_5 = 37$.

- $\gamma_4 = 0.5$ indicates a moderate negative impact of each unit increase in the tax rate on social satisfaction. This aligns with the composite effect of taxes on satisfaction (reducing income satisfaction but increasing satisfaction by reducing tourist numbers).
- capacity = 5.5 defines the competition pressure between residents and tourists in a city with limited infrastructure.
- σ controls the marginal contribution of infrastructure investment C_{infra} to social carrying capacity.
- γ_5 amplifies the negative impact of tax rate changes on social satisfaction. With $\gamma_5 = 37$, this suggests that tax rate changes significantly weaken social satisfaction, especially at higher tax rates.

The final functional relationship is expressed as:

$$\text{Residence Satisfaction} = \frac{1}{1 + e^{-\left(1 - \left(1 - \frac{0.5 N_v}{5.5 + C_{\text{infra}}}\right) - 37 T_v\right)}} \quad (8)$$

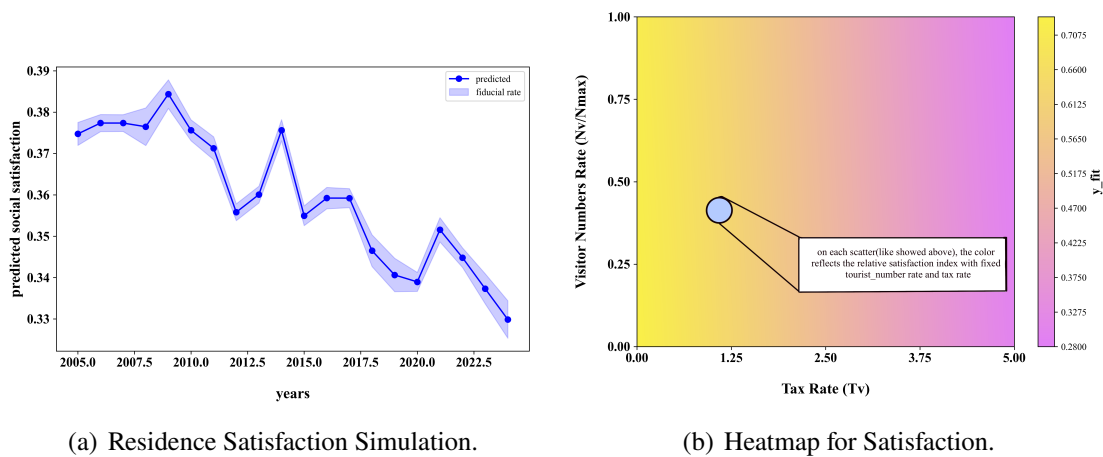


Figure 4: Fitting Accuracy Analysis(RS).

5.1.4 Another Important Evaluation Indicator

We also consider that for cities where tourism is the primary source of income, it is essential to invest the annual surplus into infrastructure maintenance and expansion. Therefore, we introduce an indicator to assess the health of infrastructure, aimed at reflecting the pressure that the growing number of tourists exerts on infrastructure.

We propose that infrastructure health can be represented by the ratio of infrastructure capacity to actual infrastructure utilization. The lower the ratio, the closer the infrastructure utilization is to saturation, indicating poorer infrastructure health. Infrastructure capacity is directly proportional to infrastructure investment, while actual infrastructure utilization is proportional to the number of tourists. Since the parameters are not critical in this case, we define the Infrastructure Health Index (IHI) as the ratio of these two factors.

$$\text{Infrastructure Health Index (IHI)} = \frac{C_{\text{infra}}}{N_v} \quad (9)$$

5.2 Policy Optimizing Model

Indicators comprehensively reflect the economic, environmental, and social impacts of tourism activities. By quantifying these effects, our model provides policymakers with concrete references to help balance the benefits of tourism against the environmental and social costs it generates.

However, due to the differences in magnitudes and units among these indicators, direct comparison and aggregation are not feasible. To ensure that all indicators carry equal weight and comparability in the sustainability assessment, we normalized each variable. This process converts each indicator into a dimensionless value within the range $[0,1]$, enabling comparisons on a consistent scale.

The normalized indicators are named as follows:

- S_1 : Economic Indicator
- S_2 : Ecological Indicator
- S_3 : Social Indicator
- S_4 : Infrastructure Indicator

Special attention was given to S_2 , which is the average of the normalized CO_2 emissions and glacier melting volumes. Next, we employed the Analytic Hierarchy Process (AHP) to allocate weights to the four sustainability indicators (S_1, S_2, S_3, S_4). AHP provides a systematic and structured method to compare the relative importance of these indicators in the context of sustainable development.

5.2.1 Weight Determination Using AHP

1. **Hierarchy Structure** In our model, the evaluation process is divided into three layers:

- **Top Layer (Goal):** The goal is to assess the sustainability of tourism, considering economic, social, and environmental factors comprehensively.

- **Criteria Layer:** This includes four key indicators: annual net income (S_1), CO₂ emissions (S_2), glacier melting volume (S_3), and the infrastructure health index (S_4).
- **Alternative Layer:** Evaluates the performance of these four indicators under different policy combinations.

2. **Construction of the Judgment Matrix** In the AHP method, weights are determined by comparing the relative importance of different indicators. To achieve this, we construct a pairwise comparison matrix that reflects the relative importance of the four indicators:

$$A = \begin{bmatrix} 1 & a_{12} & a_{13} & a_{14} \\ \frac{1}{a_{12}} & 1 & a_{23} & a_{24} \\ \frac{1}{a_{13}} & \frac{1}{a_{23}} & 1 & a_{34} \\ \frac{1}{a_{14}} & \frac{1}{a_{24}} & \frac{1}{a_{34}} & 1 \end{bmatrix} \quad (10)$$

In the judgment matrix, a_{ij} represents the relative importance of S_i and S_j . The relative urgency of the indicators is considered as follows:

- Tourism is the primary economic driver of Juneau, making economic sustainability highly significant.
- The necessity of ecological protection to mitigate glacier melting is critical.
- Resident satisfaction is somewhat conflicted and polarized, which neutralizes its overall influence.
- Infrastructure health typically requires long-term accumulation and investment to exhibit noticeable changes. Additionally, its relationship with other factors (e.g., tourist numbers and economic income) is more indirect, resulting in a lower weight.

The numerical values assigned to the pairwise comparison matrix are as follows:

$$A = \begin{bmatrix} 1 & 2 & 3 & 5 \\ \frac{1}{2} & 1 & 2 & 3 \\ \frac{1}{3} & \frac{1}{2} & 1 & 2 \\ \frac{1}{5} & \frac{1}{3} & \frac{1}{2} & 1 \end{bmatrix} \quad (11)$$

3. **Determining Indicator Weights** After constructing the judgment matrix, we calculate the weights of each indicator by solving the matrix's eigenvector through a normalization method. The specific steps are as follows:

1. Calculate the column sums of the judgment matrix:

For each column in the matrix, compute the sum of all elements.

$$A_{\text{normalized}} = \left[\frac{a_{ij}}{\text{Sum}_j} \right]$$

2. Normalize the judgment matrix:

Divide each element in a column by the sum of its respective column to obtain the normalized matrix.

3. Calculate the eigenvector:

Take the average of each row in the normalized matrix to derive the weight for each indicator:

$$w_i = \frac{1}{n} \sum_{j=1}^n a_{ij} \quad \text{for } i = 1, 2, 3, 4$$

The resulting weights for the four indicators are:

$$w_1, w_2, w_3, w_4 = [0.4824, 0.2718, 0.1575, 0.0883]$$

4. Consistency Check Since the judgment matrix is derived based on expert qualitative judgments, some inconsistencies may arise. To ensure the matrix is reasonable, a consistency check is performed. The common method is to calculate the Consistency Ratio (CR). The steps are as follows:

1. Calculate the maximum eigenvalue ($\lambda_{\max}$):

Compute the largest eigenvalue of the judgment matrix.

2. Calculate the Consistency Index (CI):

$$CI = \frac{\lambda_{\max} - n}{n - 1}$$

3. Calculate the Consistency Ratio (CR):

$$CR = \frac{CI}{RI}$$

Here, RI represents the Random Index, which depends on the size of the matrix.

From the calculations, the Consistency Ratio (CR) of our matrix is $CR = 0.002$, indicating that the matrix demonstrates excellent consistency.

Then we can derive a weighted evaluation function to assess the sustainability level of Juneau's tourism industry:

$$F = w_1 \cdot S_1 + w_2 \cdot S_2 + w_3 \cdot S_3 + w_4 \cdot S_4 \quad (12)$$

5.2.2 Optimizing Policy Combinations

Based on the characteristics and development needs of Juneau's tourism industry, we identified four major policy variables, each with three potential adjustment strategies. The table below outlines the effects achieved by each adjustment and the corresponding detailed policy descriptions.

Table 3: Policy Adjustments and Descriptions

Policy Name	Policy Content	Detailed Description
Visitor Policy	(+) N_v increase 100,000 annually	Lower entry thresholds and increase the accessibility of tourist destinations.

Continued from previous page

Policy Name	Policy Content	Detailed Description
	N_v remain constant	Improve the development and promotion of other tourist attractions to balance visitor numbers.
	$(-)N_v$ decrease 100,000 annually	Set daily visitor caps and manage tourist numbers to prevent overcrowding.
Tax Policy	$(+)T_v$ increase 0.01 annually	Levy higher taxes on tourist spending, accommodations, and activities.
	T_v remain constant	Keep the existing tax rate levels to ensure stability in the tourism market.
	$(-)T_v$ decrease 0.01 annually	Reduce taxes to encourage more tourists to visit and boost local businesses.
Environment Policy	$(+)C_{env}$ increase 5% annually	Allocate more resources to protect and restore natural ecosystems affected by tourism.
	C_{env} remain constant	Maintain current environmental investment levels to ensure sustainability.
	$(-)C_{env}$ decrease 5% annually	Reduce environmental funding, potentially increasing the risk of ecological degradation.
Infrastructure Policy	$(+)C_{infra}$ increase 10% annually	Expand infrastructure capacity to support growing tourist numbers and improve visitor experience.
	C_{infra} remain constant	Maintain current infrastructure investment to meet stable demand.
	$(-)C_{infra}$ decrease 10% annually	Reduce infrastructure spending, which may lead to decreased service quality and capacity.

To comprehensively evaluate the impact of different policy combinations on sustainable tourism in Juneau, we combined each option for the four policy variables, resulting in $3^4 = 81$ policy combinations. Each combination represents a distinct policy inclination and set of measures, encompassing comprehensive changes ranging from increasing tourist numbers and tax revenue to enhancing environmental protection and infrastructure investment.

For each policy combination, we adjusted the four fundamental variables (T_x , N_v , C_{env} , C_{infra}) and calculated the predicted values for RRR (revenue), CO₂ emissions, glacier melting, residence satisfaction, and the Infrastructure Health Index (IHI). After comparing these predicted values with historical data, we applied min-max normalization to derive the four indicators: S_1 , S_2 , S_3 , S_4 . The recursive formula is as follows:

$$\begin{aligned}
 N_v(t+1) &= N_v(t) + \Delta N_v^{\text{policy}} \\
 T_v(t+1) &= T_v(t) + \Delta T_v^{\text{policy}} \\
 C_{infra}(t+1) &= C_{infra}(t) + \Delta C_{infra}^{\text{policy}} \\
 C_{env}(t+1) &= C_{env}(t) + \Delta C_{env}^{\text{policy}}
 \end{aligned} \tag{13}$$

Subsequently, the normalized indicators were substituted into the evaluation function F , yielding the 20-year changes in F -values for all 81 policy combinations. This allows for a

comprehensive assessment of the long-term effects of each combination on the sustainability of Juneau's tourism industry.

The normalization methods are as follows:

- **For Positive Indicators:** These include RRR (revenue), Residence Satisfaction, and IHI (Infrastructure Health Index). These indicators are normalized to reflect higher values as better performance.

$$S_1 = \frac{R - \min(R)}{\max(R) - \min(R)} \quad (14)$$

$$S_3 = \frac{\text{Residence Satisfaction} - \min(\text{Residence Satisfaction})}{\max(\text{Residence Satisfaction}) - \min(\text{Residence Satisfaction})} \quad (15)$$

$$S_4 = \frac{\text{IHI} - \min(\text{IHI})}{\max(\text{IHI}) - \min(\text{IHI})} \quad (16)$$

- **For Negative Indicators:**

$$\text{CO}_2\text{normalized} = \frac{\max(\text{CO}_2) - \text{CO}_2}{\max(\text{CO}_2) - \min(\text{CO}_2)} \quad (17)$$

$$\text{Glacier}_{\text{normalized}} = \frac{\max(\text{Glacier}) - \text{Glacier}}{\max(\text{Glacier}) - \min(\text{Glacier})} \quad (18)$$

- **Ecological Indicator (S_2):** The ecological indicator S_2 is calculated as the average of the normalized CO_2 emissions and glacier melting values.

$$S_2 = \frac{1}{2}\text{CO}_2\text{normalized} + \frac{1}{2}\text{Glacier}_{\text{normalized}} \quad (19)$$

Based on the data calculations, the top ten policy combinations are shown in the following chart.

We believe that the top four policy combinations are the most valuable for reference, with their respective scores being a , b , c , and d . The specific policy combinations are displayed in the matrix:

T_v	N_v	C_{infra}	C_{env}
increase	constant	decrease	increase
constant	constant	decrease	increase
constant	increase	constant	constant
increase	constant	constant	increase
constant	constant	constant	increase

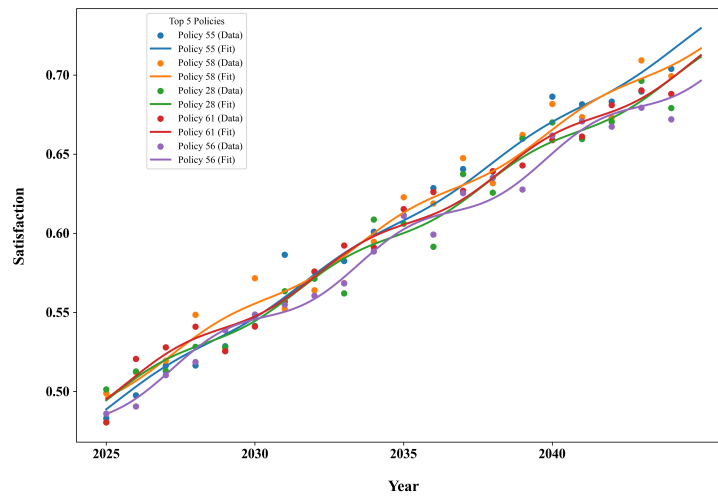


Figure 5: Optimal Sustainable Scoring Policy for the Next 20 Years(Juneau).

Based on the comprehensive evaluation of policy scores, we recommend that Juneau moderately increase taxes on tourist activities such as consumption, accommodation, and transportation to boost government revenue. At the same time, stabilizing tourist numbers through visitor diversion strategies, such as developing and promoting alternative attractions, is essential. Investment in environmental protection projects should be appropriately increased to mitigate tourism's ecological impact, while infrastructure investments, such as new commercial facilities or excessive site development, should be limited to avoid overburdening resources.

5.3 Sensitivity Analysis

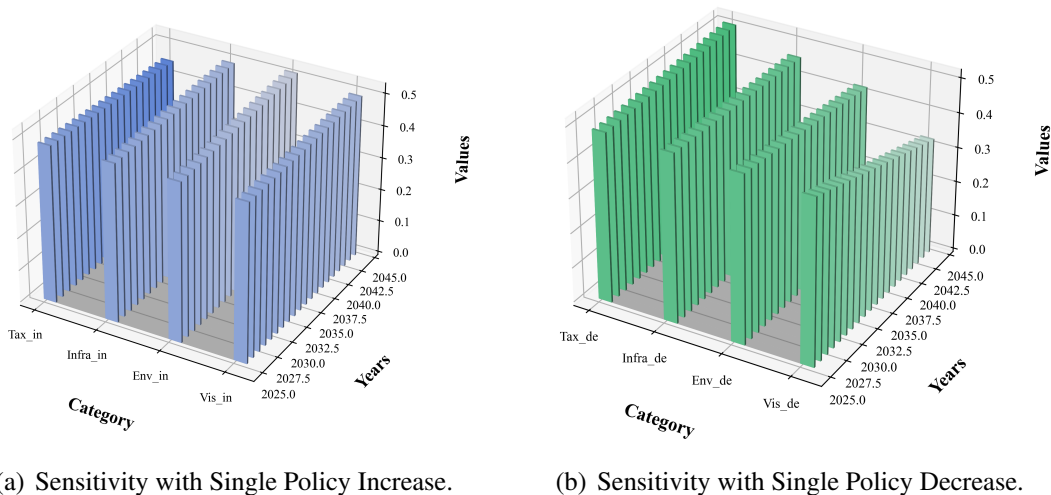


Figure 6: Sensitivity Analysis.

Through sensitivity analysis, we found that tourists are particularly sensitive to tax increases, reflecting the immediate impact of tax policies. However, due to the complex effects of tax changes on other factors, the results are difficult to predict. Limiting visitor numbers would

significantly lower sustainability scores in the long term, so we prefer adjusting environmental investment, which is more stable and offers long-term benefits.

6 Sustainable Tourism Optimization and Evaluation Model in Angkor Wat

We chose Angkor Wat as the second tourism destination because it is a world-renowned cultural heritage site facing similar overtourism challenges as Juneau. The rapid surge in tourist numbers at Angkor Wat has exerted tremendous pressure on its ecological environment, social structure, and infrastructure. Therefore, applying our sustainable tourism model to evaluate Angkor Wat's sustainability and formulating scientifically grounded policy combinations for it holds significant practical relevance and necessity.

6.1 Modified Evaluation Model

By analyzing the historical tourism data reports of Angkor Wat, we observed that the annual environmental and infrastructure investments in the region remain relatively constant. As a developing country with a lower level of regional economic development, it is reasonable to infer that the local government has limited funds. Most of the annual investments are allocated to maintaining basic infrastructure operations and ecological reinvestments, leaving little room for additional increases.

Thus, the annual environmental and infrastructure investments can be considered fixed values, and the only adjustable variables are tax rates and visitor limits.

For the four fundamental indicators, we adopt the model used in the analysis of Juneau. However, considering the distinct urban characteristics of Juneau and Angkor Wat, the ecological indicator is replaced with water pollution levels in Angkor Wat.

Furthermore, as Angkor Wat is a world cultural heritage site, its status differs significantly from natural attractions. When assessing visitor satisfaction, we referred to studies that highlighted local residents' perspectives on Angkor Wat's tourism industry. These studies suggest that resident satisfaction is strongly correlated with the level of preservation of Angkor Wat. Therefore, we approximate the annual restoration rate of Angkor Wat as a proxy for resident satisfaction.

The functional relationships for the four indicators are as follows:

$$\left\{ \begin{array}{l} R = m_1 \cdot (1 + T_v) \cdot N_v \cdot e^{-n_1 \cdot T_v^2} - C_{\text{infra}} - C_{\text{env}} \\ \text{Wastewater Production} = a_3 \cdot \left(\frac{C_{\text{env}}}{N_v} \right)^{q_1} + B_3 \\ \text{Resident Satisfaction} = \frac{1}{1 + e^{-\left(1 - \frac{\gamma_5 N_v}{\text{Capacity}_{\text{social, base}_1} + \sigma_2 C_{\text{infra}} - \gamma_6 \cdot T_v \right)}} \\ \text{Infrastructure Health Index (IHI)} = \frac{C_{\text{infra}}}{N_v} \end{array} \right. \quad (20)$$

Using nonlinear regression methods to fit the data, we determined the parameters and derived

the following parameter-free expressions:

$$\left\{ \begin{array}{l} R = 60 (1 + T_v) \cdot N_v \cdot e^{-0.01 \cdot T_v^2} - C_{\text{infra}} - C_{\text{env}} \\ \text{Wastewater Production} = 170000 \left(\frac{C_{\text{env}}}{N_v} \right)^{-1/2} + 50000 \\ \text{Resident Satisfaction} = \frac{1}{1 + e^{-\left(1 - \frac{0.5 N_v}{\text{Capacity}_{\text{social, base}_1} + C_{\text{infra}}} - 30 T_v\right)}} \\ \text{Infrastructure Health Index (IHI)} = \frac{C_{\text{infra}}}{N_v} \end{array} \right. \quad (21)$$

6.2 Policy Optimizing Model

For the two adjustable variables (T_x and R_v), we refined each variable into five adjustment options (significant increase, slight increase, no change, slight decrease, significant decrease). This allows for a more precise description and simulation of different policy decisions and their long-term impacts, resulting in $5^2 = 25$ policy combinations.

Through a process similar to the iterative and normalization steps used in the first model, we calculated the comprehensive scores for Angkor Wat over the next 20 years under different policy combinations.

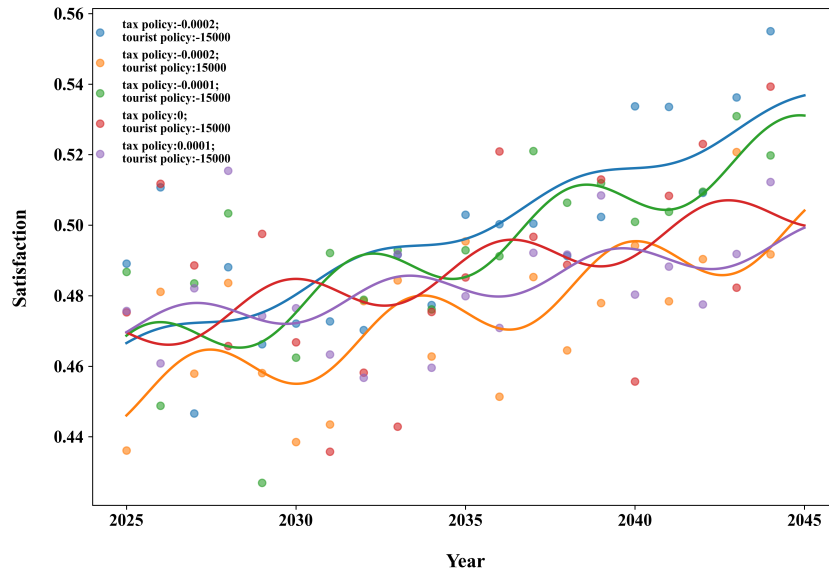


Figure 7: Optimal Sustainable Scoring Policy for the Next 20 Years(Angkor Wat).

The comparison is visualized in the heatmap below:

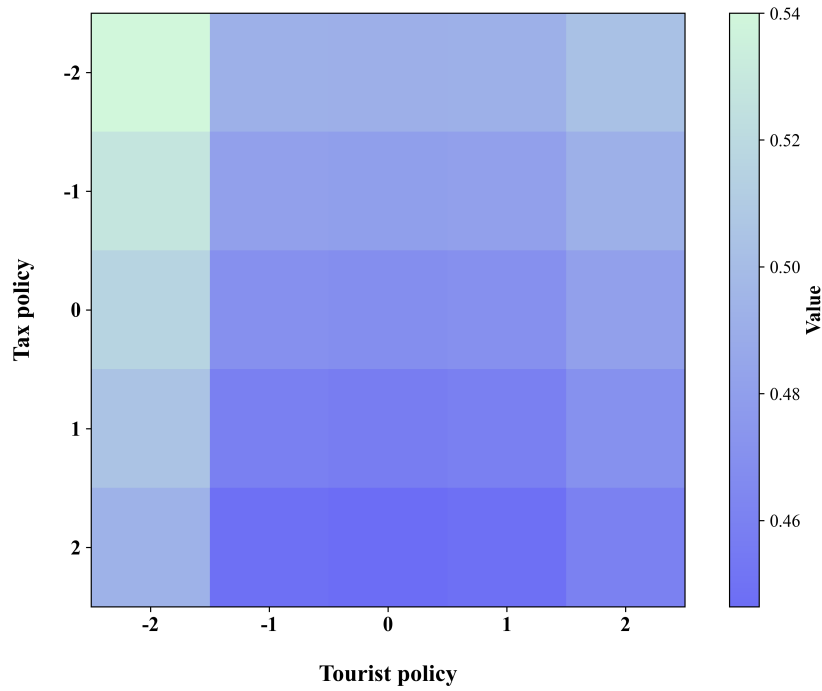


Figure 8: Policy Thermal Comparison.

It can be observed that the optimal policy combination involves reducing the annual number of tourists by 15,000 while simultaneously lowering the tax rate by 0.02%. Therefore, we suggest that for less developed regions like Angkor Wat, measures such as visitor limits should be implemented to reduce the number of tourists. At the same time, maintaining a relatively low tax rate is crucial to promote local employment and support the healthy development of the tourism industry.

7 Sustainable Tourism Optimization and Evaluation Model in Khorog

We selected Khorog in Tajikistan as a case study for promoting less-visited tourist destinations, primarily because of its unique natural landscapes and cultural heritage. At the same time, the region faces challenges of overtourism due to the ecological sensitivity of its high-altitude environment. Using our model, we can optimize tourist flows, alleviate pressure on popular destinations, and reduce the negative impacts on the environment and cultural heritage. With its infrastructure still underdeveloped, Khorog is well-suited for strategies that guide tourists, increase environmental investments, and enhance infrastructure, thereby fostering local economic and social development.

7.1 Modified Evaluation Model

Considering the underdeveloped tourism in the region and the small number of visitors far below capacity, we revised the assumptions by introducing promotional investment (P) as a new fundamental variable. The number of tourists (N_v) is modeled as

$$N_v = d \cdot \sqrt{P},$$

,reflecting a positive correlation with diminishing returns.

Given the region's high-altitude pastures, we use annual pasture degradation as the ecological indicator and rewrite the four evaluation indicators accordingly:

$$\left\{ \begin{array}{l} R = m_2 \cdot (1 + T_v) \cdot (d \cdot \sqrt{P}) \cdot e^{-n_2 \cdot (T_v)^2} - C_{\text{infra}} - C_{\text{env}} - P \\ \text{Grassland Degradation} = a_4 \cdot \left(\frac{C_{\text{env}}}{d \cdot \sqrt{P}} \right)^{q_2} + B_4 \\ \text{Resident Satisfaction} = \frac{1}{1 + e^{-\left(1 - \frac{\gamma_7 \cdot (d \cdot \sqrt{P})}{\text{Capacity}_{\text{social, base 2}} + \sigma_3 C_{\text{infra}} - \gamma_8 \cdot T_v}\right)}} \\ IHI = \frac{C_{\text{infra}}}{d \cdot \sqrt{P}} \end{array} \right. \quad (22)$$

The fitting process is omitted here for brevity.

7.2 Policy Optimizing Model

Similar to the approach used in the previous two problems, we derived the predicted F -values for the next 20 years, comparing the trends across 81 policy combinations and identifying the top five optimal policy combinations, as shown in the table below.

P	T_v	C_{infra}	C_{env}
increase	decrease	constant	increase
increase	constant	constant	increase
increase	decrease	decrease	constant
increase	decrease	decrease	increase
increase	constant	increase	constant

The model's conclusions for this type of region suggest that the government should increase P (promotional investment) alongside C_{env} to ensure ecological protection, while T_v can be slightly reduced or kept constant, and C_{infra} can be slightly increased or kept constant.

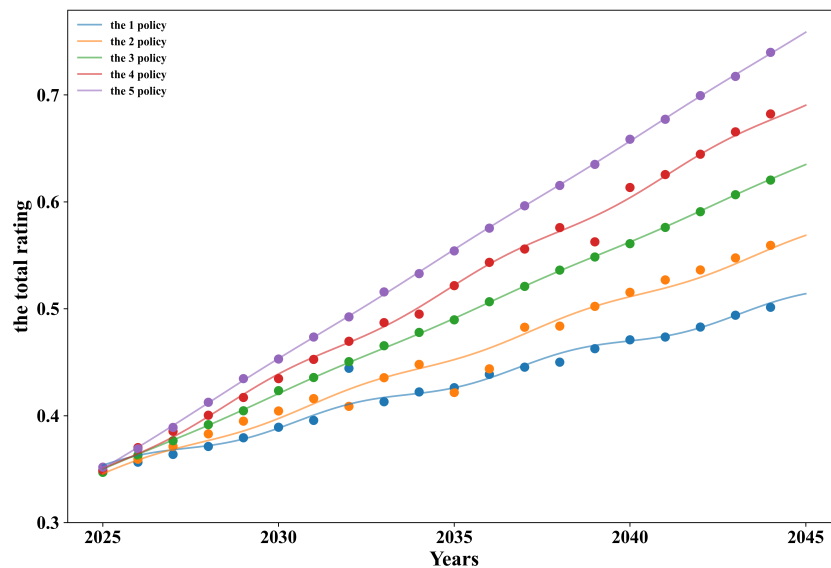


Figure 9: Optimal Sustainable Scoring Policy for the Next 20 Years(Khorog).

Increasing promotional investment P can rapidly boost the destination's visibility, attract more tourists, expand the tax base, and increase revenue. Simultaneously, increasing environmental investment C_{env} protects the fragile ecosystem and mitigates the risks of overdevelopment caused by tourist growth. Infrastructure investment C_{infra} enhances capacity and improves the visitor experience, thereby increasing overall satisfaction. Dynamic adjustments to the tax rate T_v ensure price competitiveness while providing a stable source of funding to support ecological conservation and development. This combination effectively balances economic growth and sustainability goals for less-visited destinations.

7.3 Sensitivity Analysis

We selected four policies that independently modify P , T_v , C_{env} , and C_{infra} to compare the changes in F over the next 20 years. This analysis reflects the sensitivity of the comprehensive sustainability indicator F to each individual policy measure.

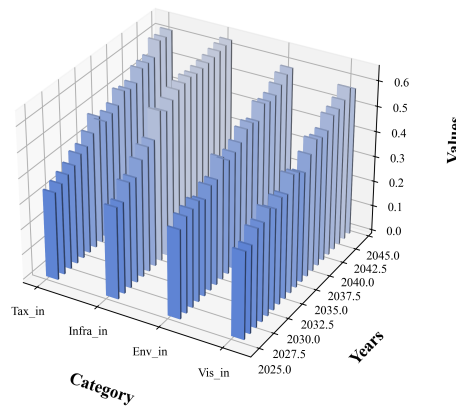


Figure 10: Sensitivity with Single Policy Shifts.

The sensitivity ranking is as follows:

$$T_v > P > C_{\text{env}} \approx C_{\text{infra}}.$$

This suggests that T_v has the highest sensitivity because it directly impacts both tourists and local residents. Promotional investment (P) exhibits an initial leverage effect but is subject to diminishing marginal returns. In contrast, the effects of C_{env} and C_{infra} are slower, longer-term, and influenced by multiple factors.

The policy implications are as follows: T_v (tax rate) is the most sensitive variable and should be adjusted cautiously to avoid significant negative impacts on tourist flow or economic revenue. Changes in P (promotional investment) can significantly boost tourist growth but must account for diminishing returns. While C_{env} (environmental investment) and C_{infra} (infrastructure investment) have lower sensitivity, they are critical for achieving long-term sustainability and should not be overlooked.

8 Strengths and Weaknesses

8.1 Strengths

- **Strength 1: Comprehensive Consideration of Key Indicators for Sustainable Tourism**
Our model evaluates economic, ecological, social, and infrastructure factors in a balanced way, reflecting the complexities of sustainable tourism. By accounting for tax rates, environmental protection, visitor numbers, and infrastructure investment, the model offers a holistic approach to managing tourism in Juneau.
- **Strength 2: Dynamic Adjustment to Policy Changes**
The model simulates 20-year trends, predicting the impact of policy changes over time. By adjusting for future variables, it provides a clear, forward-looking policy tool for decision-making.
- **Strength 5: Use of Nonlinear Regression for Parameter Estimation.**
We apply nonlinear regression to estimate parameters, capturing complex relationships between variables, such as the diminishing returns of environmental investments or tax effects on tourism.
- **Strength 4: Multi-dimensional Scoring System for Policy Evaluation**
We use scores to evaluate policies, making it easy to compare outcomes and determine the optimal policy for sustainable tourism.
- **Strength 3: Sensitivity Analysis to Guide Decision-Making**
We apply nonlinear regression to estimate parameters, capturing complex relationships between variables, such as the diminishing returns of environmental investments or tax effects on tourism.

8.2 Weaknesses

- **Weakness 1: Limited Scope of Policy Variables**
The model doesn't consider all potential factors like global economic shifts or natural disasters, which could affect the outcomes.
- **Weakness 2: Data Availability and Quality**
The model's accuracy depends on historical data quality. Incomplete or inconsistent data may lead to errors in predictions.
- **Weakness 3: Omitting the cultural impact of tourism**
To assess social satisfaction, our model excluded the cultural indicator which is hard to rate, and this may introduce some error.

Memo

To: Juneau Tourism Council

From: Team #2503068, MCM2025

Subject: Predictions and Optimization Recommendations for the Sustainability of Juneau's Tourism Industry

Date: January 28, 2025

Dear Juneau Tourism Council, our team has completed an assessment of the sustainability of Juneau's tourism industry, and based on several key indicators (annual net income, CO₂ emissions, glacier melt, resident satisfaction, and infrastructure health index), we have provided scientific predictions and optimization recommendations for policy combinations over the next 20 years.

Based on our model's predictions, We have considered different policy combinations in our evaluation, including tax rates, tourist numbers, environmental investments, and infrastructure investments. The main findings are as follows:

1. **Increasing tax rates** can boost fiscal revenue in the short term, but it may suppress tourist numbers and potentially affect the sustainable growth of the tourism industry in the long term.
2. **Increasing environmental investments** can effectively reduce CO₂ emissions and glacier melt, while improving the ecological environment.
3. **Enhancing infrastructure investment** can improve resident satisfaction and social stability, especially during peak tourist seasons.

To promote the sustainable development of tourism in Juneau, we propose the following four quantitative policy measures: First, moderately increase taxes on tourist consumption, accommodation, and transportation. We recommend raising the tax rate by 0.01% annually, which is expected to increase government revenue by 10%-15%, bringing an additional tax income of approximately \$500,000 to \$1,000,000 per year. Second, implement a tourist flow diversion strategy by developing and promoting other attractions. Third, increase investment in environmental projects, allocating an additional \$200,000 to \$300,000 per year for site protection, waste management, and pollution control. The goal is to reduce CO₂ emissions by 3%-5% annually and slow the melting of glaciers. Finally, control infrastructure development by limiting the construction of new commercial facilities and the overdevelopment of tourist sites. The annual cap for new infrastructure investment is set at \$1,000,000 to increase the maintenance budget for current facilities by 10%.

The sustainable development of Juneau's tourism requires a balance between economic benefits, ecological protection, and social stability. Combining our predictions with policies to maintain stable tourist numbers, increase environmental investments, gradually increase tax rates, and maintain infrastructure investment, Juneau is expected to achieve sustainable development of both the ecosystem and industries over the next 20 years. We recommend adjusting the policies flexibly based on the above optimization plan and conducting regular policy evaluations to ensure the best outcomes in sustainable tourism management.

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